

**MORGANTOWN, WV, USA**WMO: **724176**Lat: **39.643N**Lon: **79.916W**Elev: **1240**StdP: **14.05**Time Zone: **-5.00 (NAE)**Period: **94-19**WBAN: **13736****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 6.7        | 11.7 | -4.1                          | 4.6 | 10.0 | 0.8 | 6.0 | 14.9 | 18.0                  | 45.8 | 16.0 | 41.3 | 5.0                   | 230  | 0.417 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 18.6                   | 89.2            | 71.9 | 87.0 | 71.0 | 84.8 | 70.1 | 75.2                | 84.5 | 73.9 | 82.4 | 72.7 | 80.8 | 6.0                  | 220  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 72.4                            | 125.8 | 79.2 | 71.3 | 120.9 | 78.1 | 70.1 | 115.9 | 76.9 | 39.4          | 85.0 | 38.2 | 82.6 | 37.0 | 80.3 | 81.1              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |       |            |       |       |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|-------|------------|-------|-------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |       | n=50 years |       |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max   | Min        | Max   |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)   | (o)        | (p)   |       |     |
| (4) 15.2          | 12.8 | 11.8 | DB                         | -0.1 | 92.4               | 6.1 | 3.1                                                | -4.6 | 94.6       | -8.2 | 96.4       | -11.6 | 98.2       | -16.1 | 100.5 | (4) |
| (5)               |      |      | WB                         | -0.9 | 77.4               | 6.0 | 1.6                                                | -5.3 | 78.6       | -8.8 | 79.5       | -12.2 | 80.4       | -16.5 | 81.6  | (5) |

**Monthly Climatic Design Conditions**

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 54.0   | 31.7  | 34.8  | 42.5  | 54.1  | 62.6  | 70.3  | 73.8  | 72.7  | 66.7  | 55.6  | 45.0  | 36.8  |
|                                                                                  | DBStd       | 17.32  | 12.77 | 11.47 | 11.15 | 9.77  | 8.10  | 5.74  | 4.69  | 4.58  | 6.80  | 8.83  | 9.68  | 10.57 |
|                                                                                  | HDD50       | 2067   | 583   | 442   | 288   | 67    | 6     | 0     | 0     | 0     | 0     | 45    | 209   | 427   |
|                                                                                  | HDD65       | 4939   | 1031  | 847   | 699   | 344   | 144   | 20    | 2     | 3     | 61    | 313   | 601   | 874   |
|                                                                                  | CDD50       | 3525   | 17    | 15    | 57    | 191   | 398   | 609   | 737   | 705   | 500   | 218   | 59    | 19    |
|                                                                                  | CDD65       | 923    | 0     | 0     | 3     | 18    | 72    | 180   | 274   | 243   | 111   | 21    | 1     | 0     |
|                                                                                  | CDH74       | 7266   | 0     | 1     | 21    | 191   | 635   | 1409  | 2172  | 1827  | 843   | 160   | 7     | 0     |
|                                                                                  | CDH80       | 2100   | 0     | 0     | 2     | 33    | 153   | 403   | 706   | 545   | 234   | 24    | 0     | 0     |
| Wind                                                                             | WSAvg       | 5.0    | 6.1   | 6.1   | 6.0   | 6.0   | 4.9   | 4.4   | 3.9   | 3.6   | 3.8   | 4.5   | 5.1   | 5.5   |
| Precipitation                                                                    | PrecAvg     | 41.1   | 2.6   | 2.4   | 3.7   | 3.4   | 4.1   | 4.0   | 4.3   | 3.9   | 3.5   | 2.6   | 3.3   | 3.1   |
|                                                                                  | PrecMax     | 60.9   | 6.1   | 5.3   | 7.3   | 5.9   | 8.9   | 9.4   | 12.0  | 9.7   | 6.4   | 7.3   | 10.9  | 7.2   |
|                                                                                  | PrecMin     | 29.5   | 0.6   | 0.6   | 1.0   | 1.1   | 1.3   | 0.9   | 1.4   | 1.0   | 0.8   | 0.2   | 0.9   | 0.6   |
|                                                                                  | PrecStd     | 6.5    | 1.3   | 1.3   | 1.5   | 1.2   | 1.9   | 2.0   | 2.2   | 1.9   | 1.3   | 1.5   | 1.7   | 1.4   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 67.0  | 69.5  | 77.5  | 84.1  | 87.4  | 90.1  | 92.4  | 91.1  | 89.8  | 83.2  | 74.1  | 67.7  |
|                                                                                  |             | MCWB   | 56.4  | 54.9  | 58.9  | 61.9  | 68.2  | 72.3  | 74.4  | 73.3  | 69.8  | 67.5  | 59.9  | 58.2  |
|                                                                                  | 2%          | DB     | 62.0  | 63.3  | 71.8  | 79.4  | 84.0  | 87.2  | 89.2  | 88.3  | 86.0  | 78.4  | 69.1  | 62.5  |
|                                                                                  |             | MCWB   | 54.8  | 51.9  | 55.9  | 59.7  | 66.6  | 71.0  | 72.9  | 72.3  | 69.0  | 64.5  | 57.1  | 54.4  |
|                                                                                  | 5%          | DB     | 56.7  | 57.8  | 66.6  | 75.3  | 81.2  | 84.6  | 86.6  | 85.5  | 82.5  | 74.5  | 64.9  | 58.2  |
|                                                                                  |             | MCWB   | 51.7  | 49.6  | 53.4  | 58.4  | 65.8  | 70.2  | 71.7  | 71.3  | 68.2  | 62.3  | 54.6  | 51.4  |
|                                                                                  | 10%         | DB     | 51.1  | 52.5  | 60.9  | 70.9  | 77.4  | 82.0  | 84.1  | 83.1  | 79.1  | 70.6  | 61.2  | 54.0  |
|                                                                                  |             | MCWB   | 45.4  | 45.1  | 50.8  | 56.3  | 64.2  | 69.0  | 70.6  | 70.1  | 66.8  | 60.6  | 52.8  | 48.2  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 59.1  | 58.5  | 61.7  | 65.6  | 72.3  | 75.4  | 77.3  | 77.2  | 74.2  | 70.5  | 62.5  | 59.3  |
|                                                                                  |             | MCDB   | 64.7  | 66.0  | 74.4  | 78.3  | 82.3  | 85.7  | 87.1  | 86.1  | 82.9  | 79.0  | 70.1  | 65.2  |
|                                                                                  | 2%          | WB     | 55.5  | 54.8  | 58.1  | 62.6  | 69.4  | 73.5  | 75.3  | 75.1  | 72.1  | 67.7  | 59.3  | 56.1  |
|                                                                                  |             | MCDB   | 59.9  | 60.5  | 67.0  | 73.4  | 79.7  | 82.4  | 85.3  | 83.5  | 80.4  | 75.2  | 66.8  | 60.9  |
|                                                                                  | 5%          | WB     | 52.0  | 50.4  | 55.6  | 60.6  | 67.7  | 72.0  | 74.0  | 73.6  | 70.4  | 64.9  | 56.5  | 52.7  |
|                                                                                  |             | MCDB   | 56.3  | 56.8  | 64.2  | 70.4  | 77.2  | 81.2  | 82.7  | 81.4  | 78.0  | 71.5  | 62.6  | 56.9  |
|                                                                                  | 10%         | WB     | 45.9  | 45.8  | 52.0  | 58.7  | 65.9  | 70.7  | 72.7  | 72.3  | 68.9  | 61.8  | 53.6  | 48.2  |
|                                                                                  |             | MCDB   | 50.3  | 52.0  | 60.0  | 68.3  | 74.5  | 79.3  | 80.7  | 79.7  | 76.0  | 69.2  | 59.7  | 53.1  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 14.9  | 16.4  | 18.6  | 21.5  | 20.1  | 19.1  | 18.6  | 18.7  | 19.5  | 19.4  | 16.9  | 14.4  |
|                                                                                  |             | MCDBR  | 19.3  | 23.5  | 25.9  | 26.4  | 24.6  | 22.0  | 21.3  | 21.4  | 23.5  | 24.4  | 23.0  | 20.9  |
|                                                                                  |             | MCWBR  | 14.6  | 16.8  | 14.9  | 13.0  | 11.3  | 9.2   | 8.3   | 8.5   | 9.7   | 12.3  | 14.6  | 15.2  |
|                                                                                  | 5% WB       | MCDBR  | 18.3  | 21.8  | 21.8  | 21.6  | 20.3  | 19.0  | 18.0  | 17.7  | 18.5  | 19.4  | 19.0  | 19.0  |
|                                                                                  |             | MCWBR  | 14.8  | 17.7  | 14.7  | 12.3  | 10.1  | 8.8   | 8.0   | 8.0   | 8.8   | 11.0  | 14.1  | 15.7  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |        | 0.301 | 0.322 | 0.348 | 0.386 | 0.434 | 0.467 | 0.488 | 0.498 | 0.411 | 0.365 | 0.324 | 0.301 |
|                                                                                  | taud        |        | 2.459 | 2.384 | 2.388 | 2.314 | 2.187 | 2.127 | 2.087 | 2.031 | 2.267 | 2.398 | 2.488 | 2.496 |
|                                                                                  | Ebn at Noon |        | 276   | 285   | 288   | 282   | 269   | 259   | 252   | 246   | 263   | 265   | 265   | 267   |
|                                                                                  | Edh at Noon |        | 25    | 31    | 35    | 40    | 46    | 49    | 51    | 52    | 39    | 30    | 24    | 23    |
| All-Sky Solar<br>Radiation                                                       | RadAvg      |        | 508   | 768   | 1093  | 1460  | 1664  | 1833  | 1792  | 1643  | 1360  | 944   | 649   | 443   |
|                                                                                  | RadStd      |        | 57    | 120   | 92    | 131   | 143   | 139   | 109   | 95    | 151   | 110   | 79    | 49    |

**Historical Trends**

|                         |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                         |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                         |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Station Only            |  | +0.83 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +263  | +153  |
| Regional (29 neighbors) |  | +0.68 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +205  | N/A   |

Nomenclature: See separate page